EX NO : 7

DATE : 28/8/26

FLOW CONTROL USING SLIDING WINDOW PROTOCOL

AIM:

To Implement flow control at the Data Link Layer using the Sliding Window Protocol and simulate the flow of frames from one node to another using acknowledgements.

ALGORITHM:

Sender Side

1) Start the program
2) Read the window size from the user.
3) Read the text message from the user.
4) Divide the message into frames, considering one character per frame
5) Assign a frame number to each character.
6) Create frame with the fields Frame.NO and DATA
7) Send a set of frames according to the window size.
8) Store the transmitted frames in Sender_Buffer.txt
9) Wait for the receiver's acknowledgment.
10) Read the acknowledgement from Receiver_Buffer.txt.
11) If the correct ACK is received, Slide the window and send the next set of frames.
12) If a NACK is received, resend the required frames.
13) Continue until all frames are transmitted.
14) Stop the program.

sender

```
import time
import os
window_size = int(input("Enter Window Size: "))
message = input("Enter Text Message: ")
frames = []
for i, char in enumerate(message, start=1):
    frames.append((i, char))
total_frames = len(frames)
base = 0
print("\nTotal Frames:", total_frames)
while base < total_frames:
    window = frames[base:base + window_size]
    print("\nSending Frames:")
    if os.path.exists("Receiver_Buffer.txt"):
        os.remove("Receiver_Buffer.txt")
    with open("Sender_Buffer.txt", "w") as file:
        for frame_no, data in window:
            frame = f"[Frame No: {frame_no}, DATA: {data}]"
            print(frame)
            file.write(frame + "\n")
    print("\nWaiting for acknowledgement...")
    while not os.path.exists("Receiver_Buffer.txt"):
        time.sleep(0.5)
    with open("Receiver_Buffer.txt", "r") as file:
        response = file.read().strip()
    print("Receiver Response:", response)
    if response.startswith("ACK"):
        ack_no = int(response.split(":")[1].strip())
        expected_ack = base + len(window)
        if ack_no == expected_ack:
            print("Correct ACK received.")
            base = ack_no
        else:
            print("Unexpected ACK. Resending current window.")
    elif response.startswith("NACK"):
        nack_no = int(response.split(":")[1].strip())
        print("NACK received for Frame:", nack_no)
        base = nack_no - 1
print("\nAll frames transmitted successfully.")
print("Message:", message)
```

receiver

```
import time
import os
expected_frame = 1
last_buffer = ""
print("Receiver started.")
while True:
    if not os.path.exists("Sender_Buffer.txt"):
        time.sleep(0.5)
        continue
    with open("Sender_Buffer.txt", "r") as file:
        data = file.read()
    if data == last_buffer:
        time.sleep(0.5)
        continue
    last_buffer = data
    lines = data.strip().splitlines()
    if not lines:
        continue
    print("\nReceived Frames:")
    error = False
    last_received = expected_frame - 1
    for line in lines:
        print(line)
        start = line.find("Frame No:") + len("Frame No:")
        end = line.find(",")
        frame_no = int(line[start:end].strip())
        if frame_no == expected_frame:
            last_received = frame_no
            expected_frame += 1
        else:
            print("Unexpected Frame Number:", frame_no)
            with open("Receiver_Buffer.txt", "w") as file:
                file.write(f"NACK: {expected_frame}")
            print("NACK sent:", expected_frame)
            error = True
            break
    if not error:
        with open("Receiver_Buffer.txt", "w") as file:
            file.write(f"ACK: {last_received}")
        print("ACK sent:", last_received)
    time.sleep(0.5)
```

```
Problems  Output  Debug Console  Terminal  Ports

PS D:\SlidingWindowLab> python receiver.py
Receiver started.

Received Frames:
[Frame No: 1, DATA: H]
[Frame No: 2, DATA: E]
[Frame No: 3, DATA: L]
ACK sent: 3

Received Frames:
[Frame No: 4, DATA: L]
[Frame No: 5, DATA: O]
ACK sent: 5
```

```
Problems  Output  Debug Console  Terminal  Ports

PS D:\SlidingWindowLab> python sender.py
Enter Window Size: 3
Enter Text Message: HELLO

Total Frames: 5

Sending Frames:
[Frame No: 1, DATA: H]
[Frame No: 2, DATA: E]
[Frame No: 3, DATA: L]

Waiting for acknowledgement...
Receiver Response: ACK: 3
Correct ACK received.

Sending Frames:
[Frame No: 4, DATA: L]
[Frame No: 5, DATA: O]

Waiting for acknowledgement...
Receiver Response: ACK: 5
Correct ACK received.

All frames transmitted successfully.
Message: HELLO
```

Receiver Side:

1) Start the program
2) Read the frames from Sender_Buffer.txt
3) Check the frame numbers.
4) Compare the received frame number with the expected frame number.
5) If the frame number is correct, generate the appropriate ACK.
6) If the frame number is Incorrect, generate a NACK.
7) Store the ACK/NACK in Receiver_Buffer.txt.
8) Continue until all frames are received.
9) Stop the program.

RESULT:

Thus the Sliding Window Protocol using Flow Control was Implemented and executed Successfully.